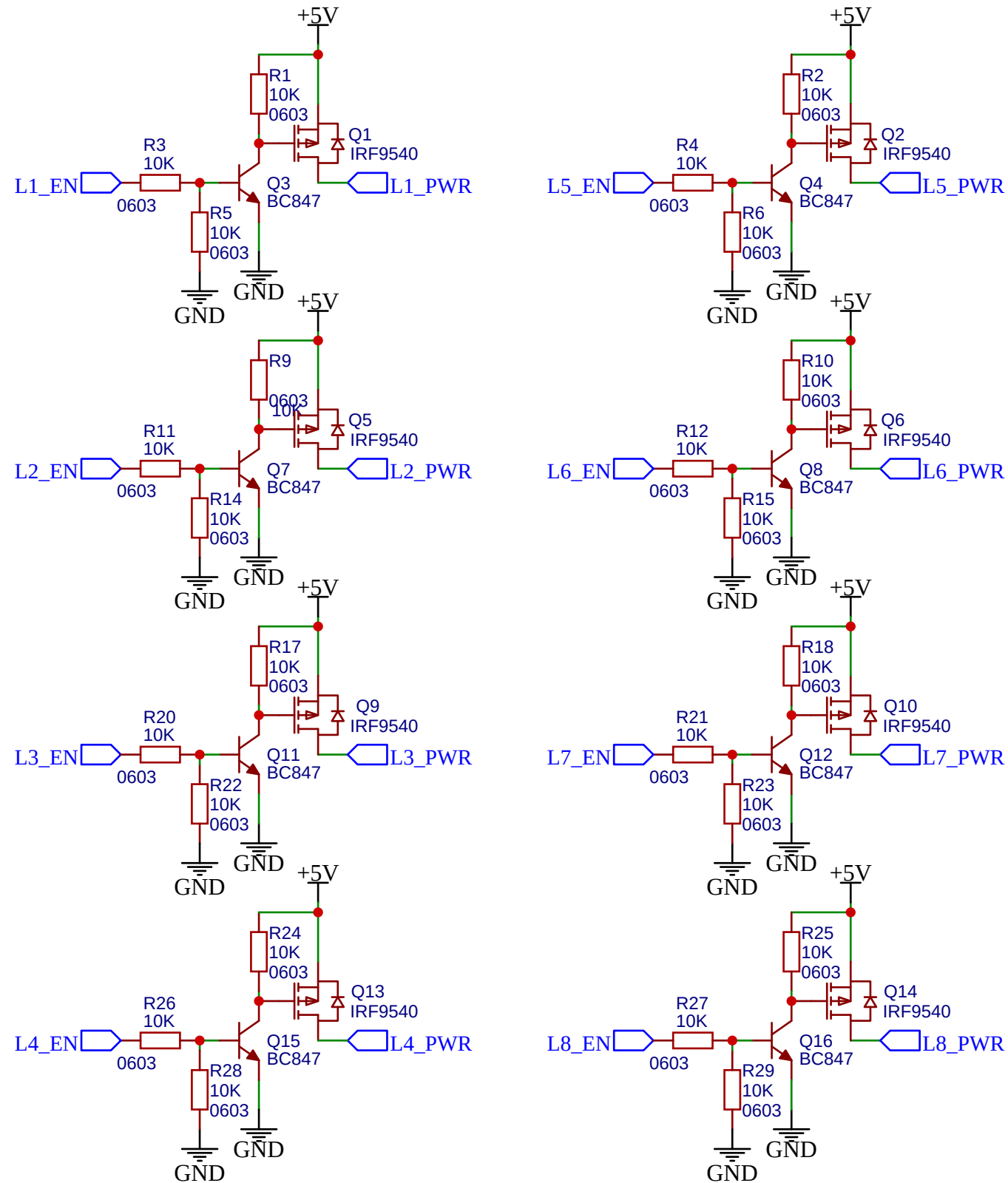
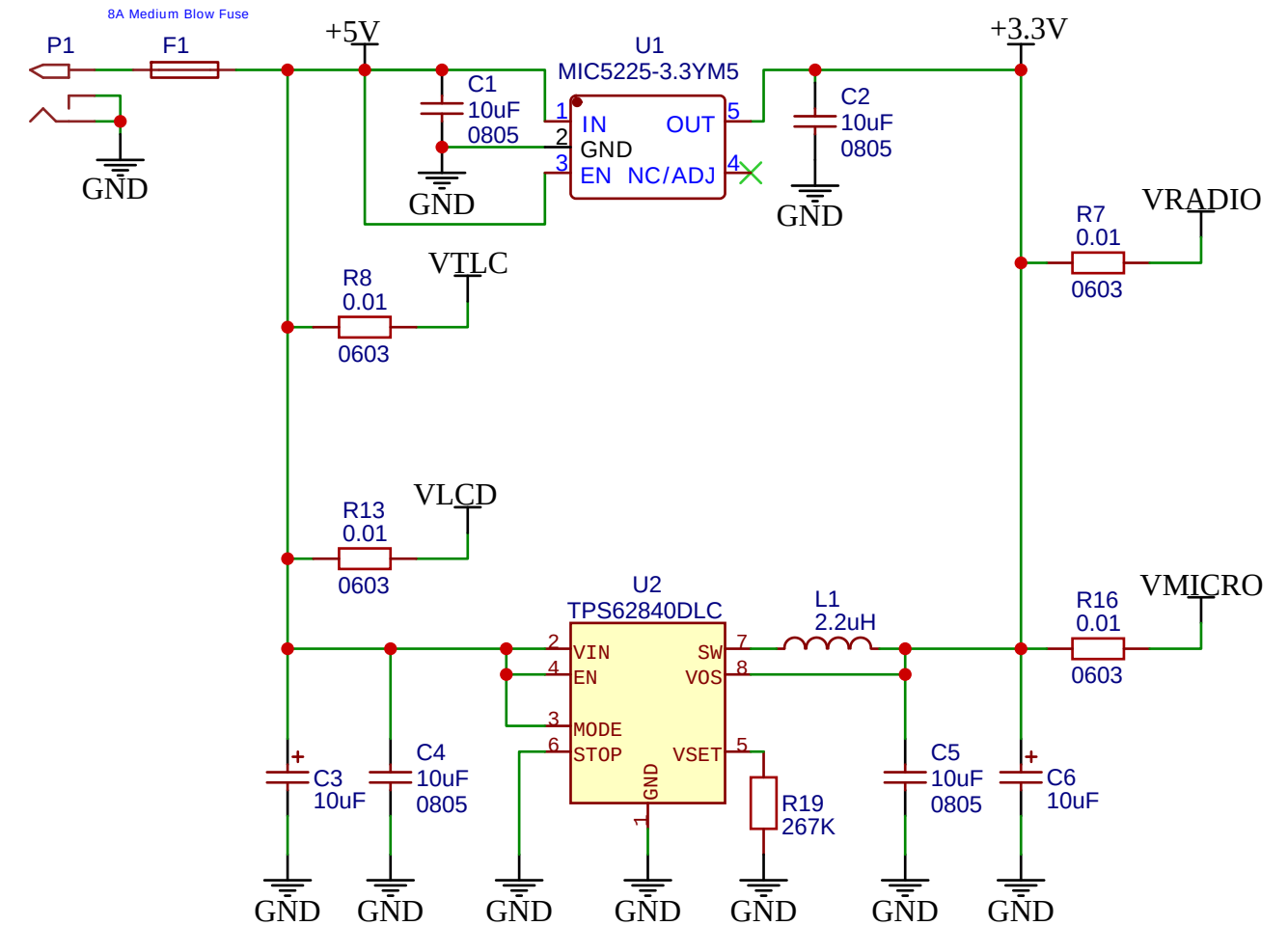


LED Power Load Switches



Power Supply



5V 5-10A PSU recommended.

64 RGB leds * 3 * 20mA = 3.84A max current

Each of these load switches powers one 8x8 layer of RGB LEDs designed for 20mA max per R/G/B LED.

The 8 layers of 8x8 RGB LEDs are duty cycled using persistence of vision to create the illusion of a fully lit cube.

TITLE:

Power Supply & Power Load Switches

REV: 1.0

EasyEDA

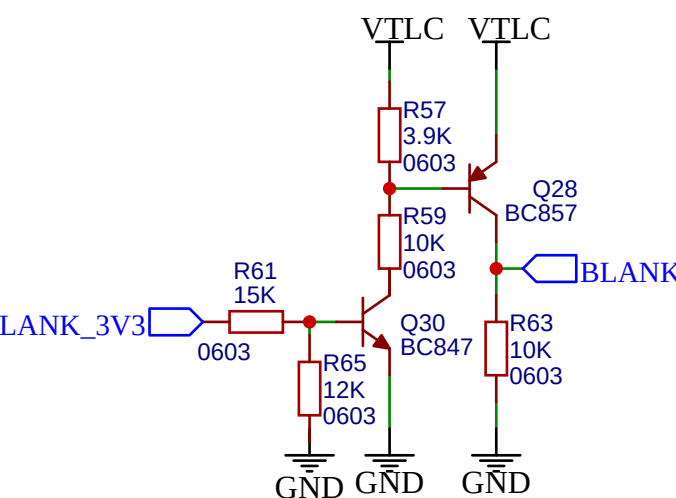
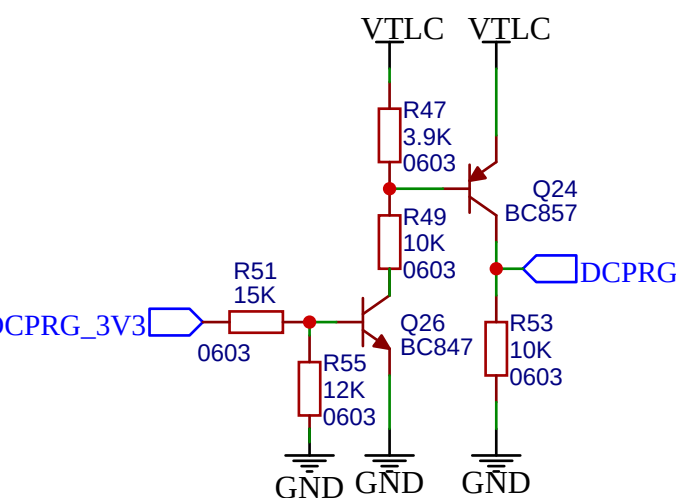
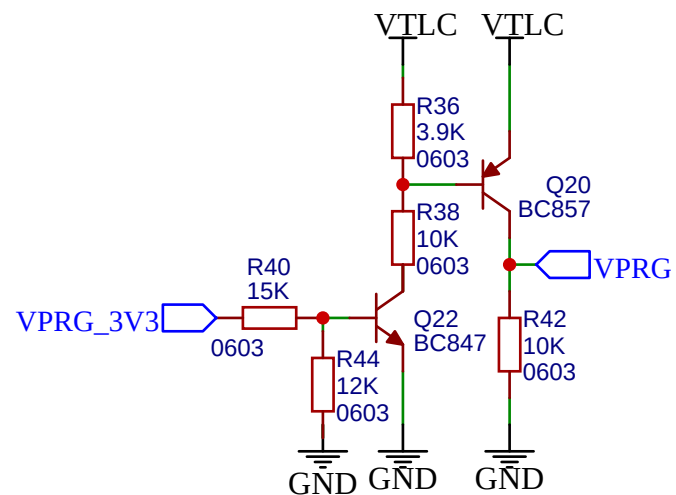
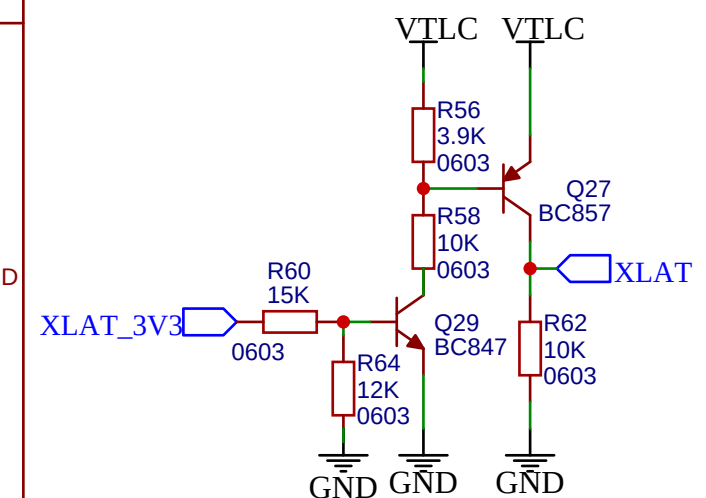
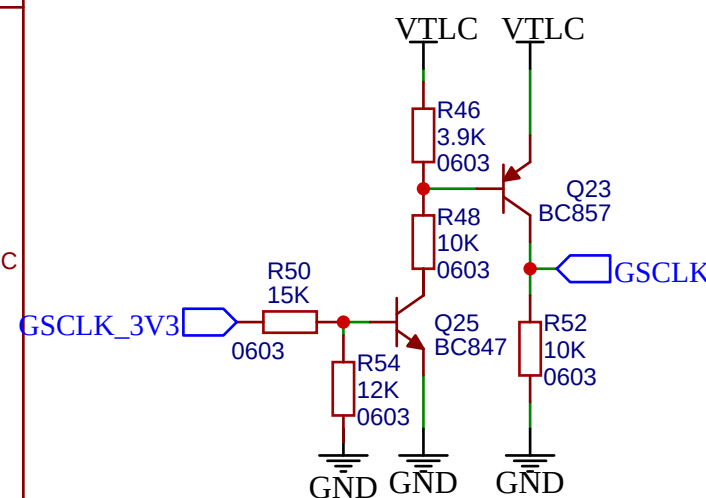
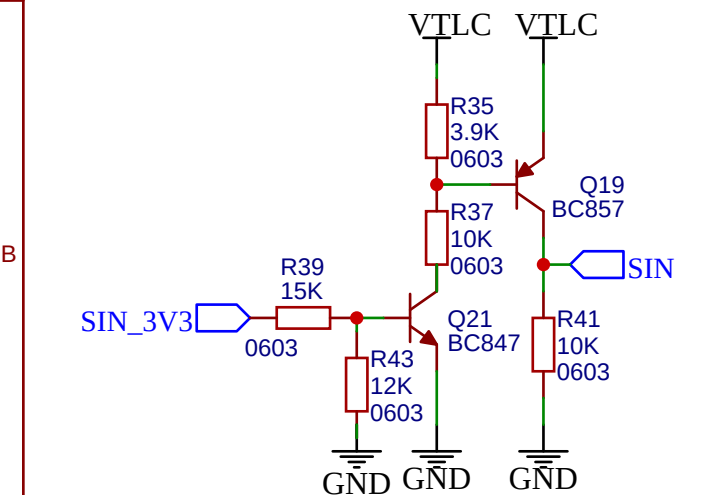
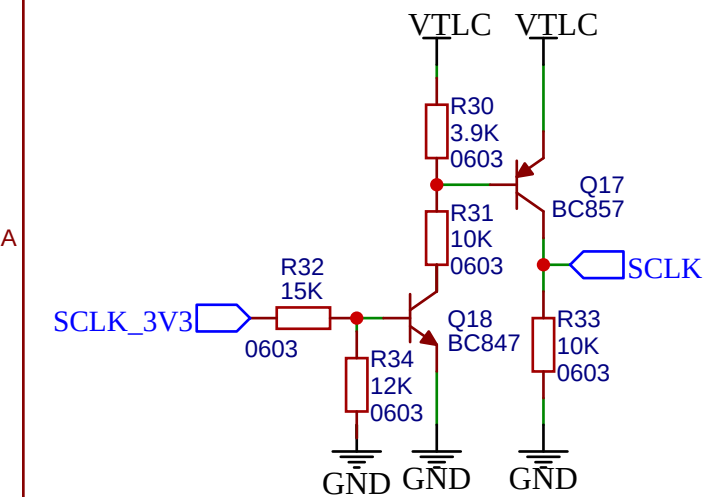
Company:

Date: 2019

Drawn By: joelamichaelis

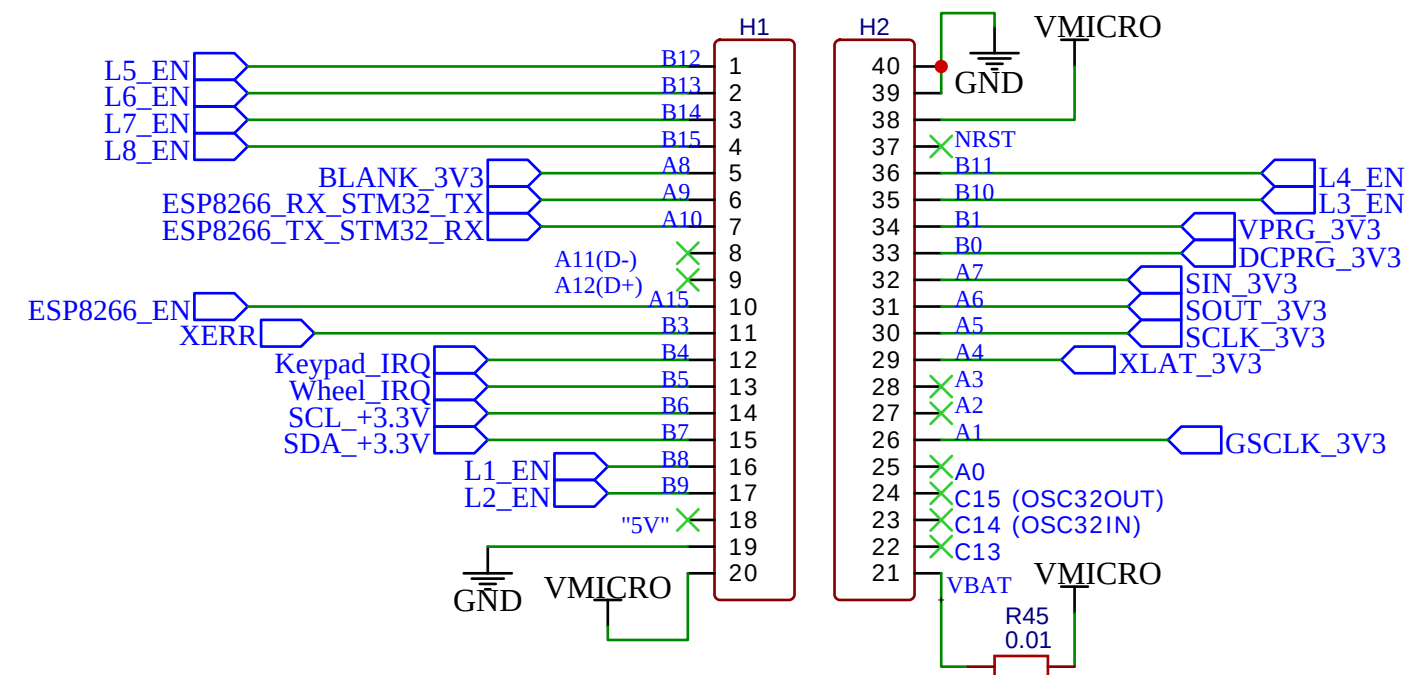
Sheet: 1

PWM Driver - Micro Level Shifters



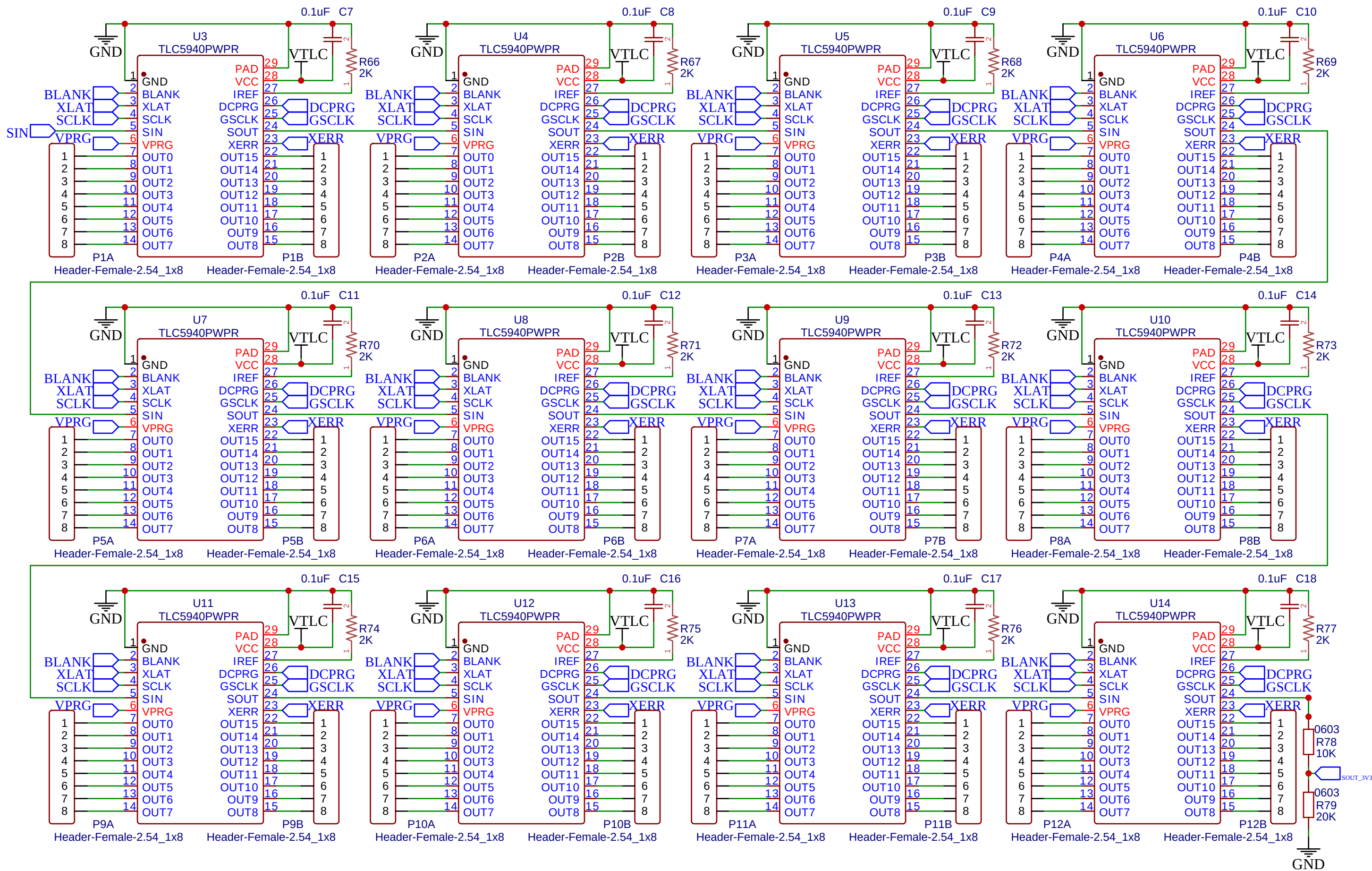
STM32F103 Blue Pill Adapter

micro headers are soldered on the bottom of the pcb despite what the pcba render shows.



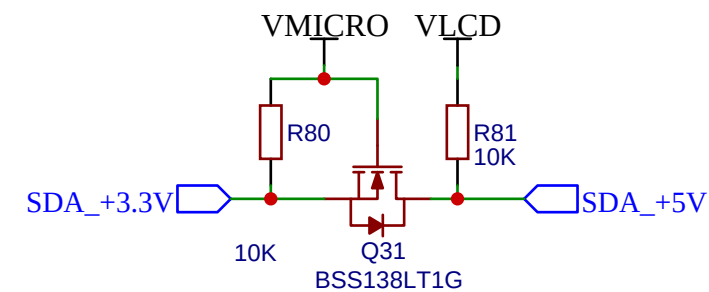
Note: In retrospect the way I implemented this 3.3V - 5V level shifting was suboptimal. A couple of the high speed signals switch faster than the bandwidth these level shifters can support. I remember learning after ordering and building this rev1 that a txs0108 level shifter IC is far superior. Trying to recall if I made a rev2 and ordered pcbs to address this issue or simply reworked the rev1.

TITLE: Micro & Level Shifters		REV: 1.0
EasyEDA	Company:	Sheet: 1/1
	Date: 2019	Drawn By: joelamichaelis

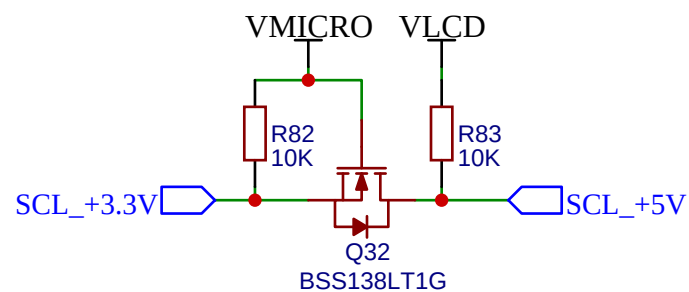
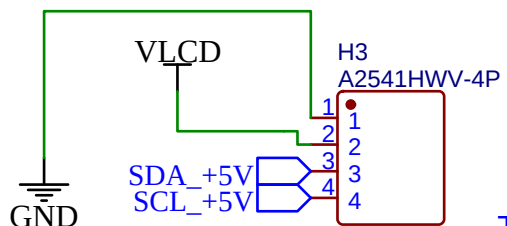


The 2k resistors set the current limit per channel

TITLE: PWM Drivers		REV: 1.0
EasyEDA	Company:	Sheet: 1/1
	Date: 2019	Drawn By: joelamichaelis

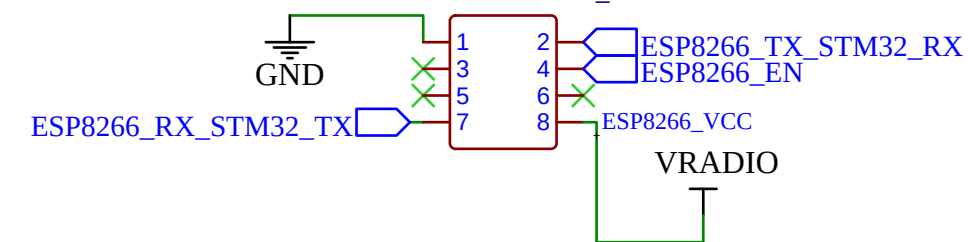


LCD ADDR = 0x27



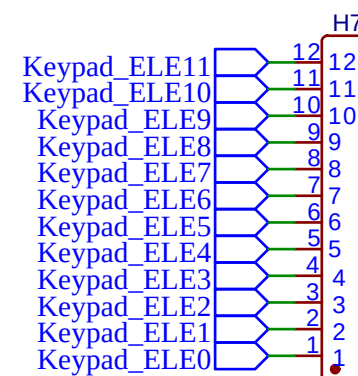
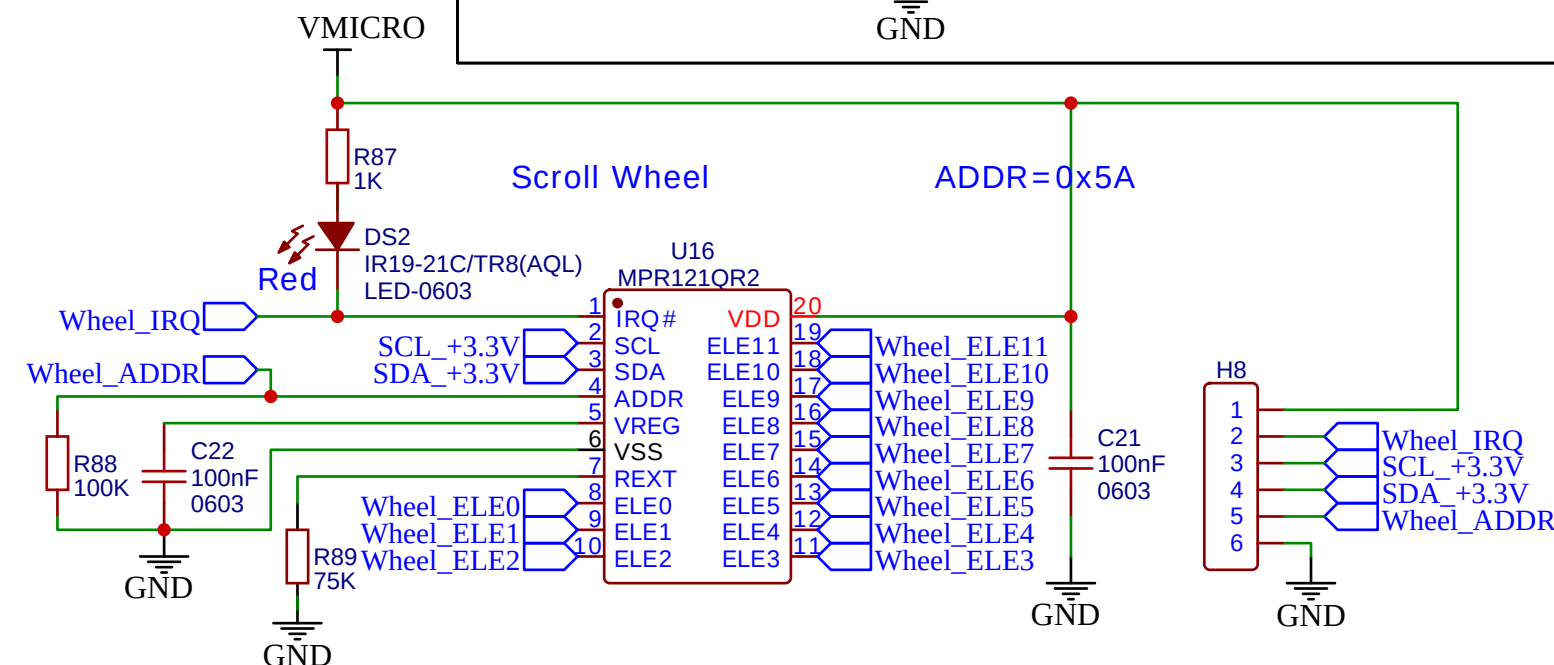
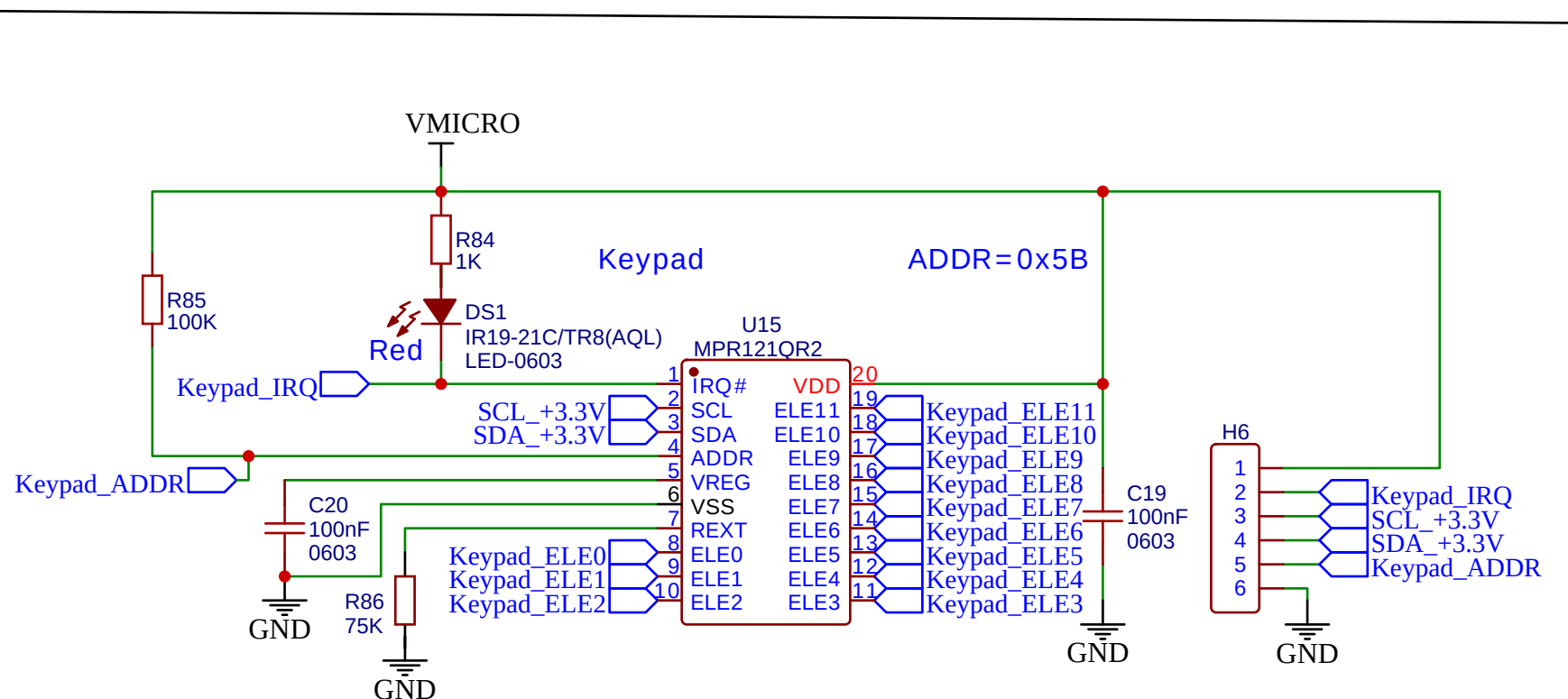
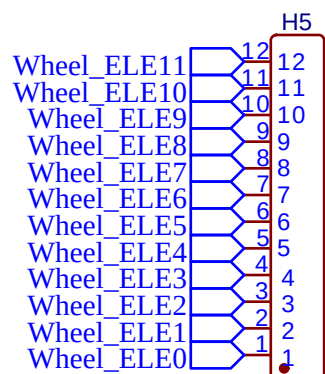
ESP8266 Connector

H4
Pins 3&5 are optional GPIO
Header-Female-2.54_2x4



Intent was to use this as an RF comms chip to remotely control the cube.
Never actually developed any functionality with it.

The peripherals consist of a LCD front screen, and a capacitive touch keypad/scrollwheel. There's also an ESP8266 rf module which was never really used.



TITLE: Peripherals		REV: 1.0
EasyEDA	Company:	Sheet: 1/1
	Date: 2019	Drawn By: joelamichaelis